

EXERCISES



PART A – COMPETENCY FOCUSED QUESTIONS

I. Multiple Choice Questions:

Tick (✓) the correct answer:

1.

int	nextInt()
short	nextShort()
double	nextDouble()
float	nextFloat()
char	next().charAt()
string	nextLine()

Which of the following Java packages contain these methods?

- (a) java input.output.* (b) java.io.*
(c) java.sc.util.* (d) java util.*
2. Which version of JDK does not contain Scanner class?
(a) JDK 1.5 (b) JDK 1.6 (c) JDK 1.2 (d) None of these
3. Which method of Scanner class is not used to accept integer data?
(a) nextInt() (b) nextByte() (c) nextLong() (d) next()
4. The function boolean hasNext() is used to:
(a) accept an integer data (b) accept either true or false
(c) accept another token (d) accept a symbol
5. Which of the following functions accept a line of text through the Scanner object?
(a) nextLine() (b) next()
(c) nextBoolean() (d) nextShort()
6. What function is used to close the Scanner class?
(a) close() (b) exit() (c) break() (d) stop()
7. Read the following text and choose the correct answer:
The Scanner class basically works on the principle of Tokens in the set of input values. It is a series of characters that ends with a delimiter.
Which of the following special characters used as the default delimiter?
(a) Colon (b) Semi colon
(c) White space (d) Exclamation

II. Fill in the blanks:

- JDK 1.5 allows a special class to input from the console. This class is termed as class.
- package is necessary to be imported in order to use Scanner class.
- Token is a set of characters separated by

4. The default delimiter used in Scanner object is _____.
5. System.in receives the input from the _____ for the Scanner object.
6. Method _____ can be used to accept a token from the Scanner object.
7. _____ method reads a token as an integer.
8. The method which checks whether the next token of the Scanner object is a floating type value or not is _____.
9. _____ is the method used to check whether the next token is a boolean type value or not.
10. _____ number of objects can be input to a Scanner object.

III. Write TRUE or FALSE:

1. Scanner class is a useful package of JDK 1.5.
2. Strings can even be input without using Scanner object.
3. The word 'new' is a keyword to create any object.
4. You need not be aware about the number of tokens to be input to a Scanner object.
5. You can terminate a string even by using carriage return.
6. nextFloat() can also accept an integer token.
7. hasNextInt() results in true, if the next token can be interpreted as an integer.
8. nextInt() can be used to accept an exponential value.

IV. Assertion and Reasoning based questions:

Assertion (A): Scanner class is available in the system package java.util.*.

Reason (R): You must import java.io as well as java.util package to avail the facilities contained in the Scanner class.

Based on the above discussion, choose an appropriate statement from the options given below:

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not a correct explanation of Assertion(A).
- (c) Assertion (A) is true and Reason (R) is false.
- (d) Assertion (A) is false and Reason (R) is true.

V. Application based questions:

In a quiz competition, a program is shown on the LCD screen to accept data of different types using Scanner class and find the frequency of integer, real and long type values. However, the program is incomplete having some places marked blank with ?1?, ?2?, ?3? and ?4? to be filled with appropriate function/ expression. The incomplete program is as given below:

```
import java.util...?1?...;
class Sample
```

```

{
public static void main(String args[] )
{
    int c=0, d=0, e=0;
    Scanner in = new Scanner(.....?2?.....);
    System.out.println("Enter data values ending with dot:");
    while(true)
    {
        if(in.hasNext( ) == '.')
            break;
        if(in.hasNextInt( ))
            c++;
        if(in.hasNextDouble( ))
            .....?3?.....;
        if(in.hasNextLong( ))
            e++;
    }
    System.out.println("Number of integer data"+.....?4?.....);
    System.out.println("Number of real values"+d);
    System.out.println("Number of long type data"+e);
}
}

```

With reference to the above program, answer the following questions:

- What class name should be filled in place of ?1?
- Name the parameter to be filled in place of ?2? that represents standard input device (keyboard).
- Write the statement in place of ?3?
- What value should be filled in place of ?4?

VI. Unsolved Java Programs:

- Using the Scanner class, write a program to input temperatures recorded in different cities in °F (Fahrenheit). Convert and display each temperature in °C (Celsius). The program terminates when user enters a non-numeric character.
- Write a program to accept a set of 50 integers. Find and display the greatest and the smallest numbers by using the Scanner class method.
- Write a program (using Scanner class) to generate a pattern of a token in the form of a triangle or in the form of an inverted triangle depending upon the user's choice.

[ICSE MODEL]

Sample Input: Enter your choice 1 Enter your choice 2

Sample Output:

*	* * * * *
* *	* * * *
* * *	* * *
* * * *	* *
* * * * *	*

- In a competitive examination, a set of 'N' number of questions results in 'True' or 'False'. Write a program by using Scanner class to accept the answers. Display the frequency of 'True' and 'False'.

5. Write a program to accept a sentence in mixed case. Find the frequency of vowels in each word and display the words along with their frequencies in separate lines.

Sample Input: We are learning Scanner class

Sample Output:

Word	Frequency of vowels
We	1
are	2
learning	3
Scanner	2
class	1

6. Write a program by using Scanner class to input a sentence. Display the longest word along with the number of characters in it. [ICSE 2009]

Sample Input: We are learning Scanner class in Java

Sample Output: The longest word: learning
Number of characters: 8

7. Write a program by using Scanner class to input a sentence. Display each word of the sentence along with the sum of the ASCII codes of its characters.

8. Write a program by using Scanner class to accept a set of positive and negative numbers randomly. Display all the negative numbers first and then all the positive numbers without changing the order of the numbers.

Sample Input: 1, 5, -2, 6, -3, -5, 8, -4, 9, 12

Sample Output: -2, -3, -5, -4, 1, 5, 6, 8, 9, 12

9. Write a program to generate and display random numbers between 1 to N by using Scanner class taking N from the console.

10. Consider a string:

THE COLD WATER AND THE HOT WATER GLASSES ARE KEPT ON THE TABLE

Write a program by using Scanner class to enter the string and display the new string after removing the repeating token 'THE'. The new string is:

COLD WATER AND HOT WATER GLASSES ARE KEPT ON TABLE

11. Using Scanner class, write a program to input a string and display all the tokens of the string which begin with a capital letter and end with a small letter.

Sample Input: The capital of India is New Delhi

Sample Output: The India New Delhi

12. Write a program to input a string by using Scanner class. Display the new string which is formed by the first character of each string.

Sample Input: Automated Teller Machine

Sample Output: ATM

13. Consider the following statement:

"26 January is celebrated as the Republic Day of India"

Write a program (using Scanner class) to change 26 to 15; January to August; Republic to Independence and finally display as:

"15 August is celebrated as the Independence Day of India" [ICSE 2006]

14. Write a program by using Scanner class to input principal (P), rate (R) and time (T). Calculate and display the amount and compound interest.

Use the formula: $A = P \left(1 + \frac{R}{100} \right)^T$

$$CI = A - P$$

PART B - SUBJECTIVE QUESTIONS

I. Differentiate between the following:

1. Scanner object and BufferedReader object
2. nextFloat() and nextDouble()
3. next() and nextLine()
4. hasNext() and hasNextLine()
5. hasNextInt() and hasNextBoolean()

II. Answer the following:

1. What do you mean by the Scanner class?
2. Write syntax along with an example to create a Scanner object.
3. Mention the syntax to use '?' as a token delimiter.
4. Write down the use of nextInt() method.
5. What is the application of hasNextBoolean() method?
6. Name the package which can be imported to use Scanner class.
7. In what way can the following data be read from the Scanner object?
(a) Integer type (b) Float type (c) Double type (d) String type (e) Boolean type